

Inter-Rater Reliability Training

Blind match scoring rubric for teacher advice against model notes

This document trains coders to apply one rubric in the same way. A second coder, blind to which note is the true note, scores each teacher's independent advice against both notes. Reliability is then checked by comparing two coders on the same items.

1. Definition of the rubric

1.1 The three-point scale

Work through the questions in this order and stop at the first yes.

1. Does the advice name the same feature as the note, or a clear synonym? If yes, score 2.
2. If not, do both point in the same general direction? If yes, score 1.
3. Otherwise, score 0. This includes advice that opposes the note.

Score	Label	Definition	Ask yourself
2	Match	The teacher's advice names the same feature or a clear synonym	Would the teacher say these two are the same suggestion in different words?
1	Partial	Same general direction but different specific feature	Do they aim at the same thing but propose different actions?
0	No match	Unrelated or contradictory.	Do they aim at different things, or do they oppose each other?

1.2 What is being scored

Every score compares one piece of teacher advice with one note. A teacher's advice therefore receives two scores, one for each note.

Term	Meaning
Advice	What the teacher wrote on their own in the feedback, before seeing either note, about what to change or do in the lesson segment.
Note	One of the two recommendations shown for the same segment. One is the model's true note and one is the foil. The coder does not know which.
Feature	The specific teaching action or focus that a piece of advice or a note names, for example recall of prior knowledge, drawing concept maps, or pair discussion.
Direction	The broader aim a feature serves, for example retention of facts, student involvement, or linking verbal and visual instruction. Different features can share a direction.

2. Rules with examples

Clip A - Expert teacher advice

Clip A

1. In the clip shown, is there anything the teacher is doing well?

- * Asking questions
- * Involving students' responses

2. In the clip shown, is there anything that the teacher is not doing well?

- * No well structured questions
- * Changed the framing of questions after receiving undesired answers
- * No visual / demonstration / activity to show the effect of indicators.

3. Suggest, if there is any scope of improvement

- * Visuals / ppt / Activity / Demonstration should be included
- * Questions should be well framed based on above activities / demonstration
- * Questions should be in inductive - deductive format so that students should achieve the generalisation at their own

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Clip B

1. In the clip shown, is there anything the teacher is doing well?

- * Good knowledge of the content
- * Good voice modulation and confidence
- * Taking up the doubts of students

2. In the clip shown, is there anything that the teacher is not doing well?

- * All the students were not involved so it was a bit noisy
- * Some visuals must be involved.
- * Topic/content should be developed with the students' involvement
- * Not able to pick the interest of all the students

3. Suggest, if there is any scope of improvement

- * Any model / demonstration / Diagram / ppt. should be there in support of the content.
- * Students should be involved in content

1a). The teacher should ask the students to recall and name the examples of indicators multiple times; this enables better retention due to constant repetition. **(Foil recommendation).**

1b). The teacher should avoid asking the students to recall the name of the synthetic indicator which promotes retention over understanding of the concept. **(True recommendation).**

The foil is recommending asking students question with the intention of recommendation. Here, the foil is a partial match as the intention of asking questions is correct the features are different as the foil talks about retention question, where the expert teacher advises the question should be inductive-deductive format.

Since it is not clear whether the retention question is answered or not, it is a partial match.

3a). The teacher instructs the students to passively write down definitions. The teacher can instead ask the students to write down their understanding of the concept of indicators. **(True recommendation).**

3b). The teacher should encourage thorough note-taking in the class, as writing down the examples and types of indicators, the students create a concise document for revision of the concepts they learn in class. **(Foil recommendation).**

Since the expert teacher has not explicitly mentioned the effects of jotting down as a positive or negative influence on learning. Both the recommendations should given zero.

Clip B - Expert teacher advice

2a). Throughout the class, teacher could have given relatable analogies and examples to easily relate and visualise the concept of 'activation of deoxyribonucleotides. **(True recommendation).**

2b). The teacher should spend more time to establish the precise definition, terminologies and mechanisms of deoxyribonucleotide activation concept this will help in further retention. **(Foil recommendation).**

According to expert teacher advice, they are suggesting to use visuals like models, demonstrations, flashcards, and PPTs to support the content and involve more students. While scoring 2A and 2B, we should keep in mind that although the teacher is suggesting mechanisms to promote visualisation of the concept. The true recommendation card states different mechanisms, which are 'providing analogies and examples'. Hence, this comes under partial match, as the general direction is the same, but a different specific feature is present from the true card. Hence, the true card recommendation should be encoded as 1 rather than 2.

3a). The teacher shifting from verbal explanation to diagrams without enough verbal framing will confuse the children and their attention might drop. **(Foil recommendation).**

3b). One thing the teacher should continue doing is, when the teacher drew the diagram of conversion of AMP to ATP, it helped the children's retention of the information by converting a complex process into an equation. **(True recommendation).**

The mechanism explained in the true recommendation card is converting word information into an equation (changing the mode of information), which is close to Bloom's understanding (interpretation). While the explanation given by the expert teacher talks about mechanisms involving visualisations like models, demonstrations, flashcards, and PPTs to support the content and involve more students. Hence, this comes under partial match, as the general direction is the same, but a different specific feature is present from the true card. Hence, the true card recommendation should be encoded as 1 rather than 2.